Ethanol Blending in Gasoline - Argentina

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

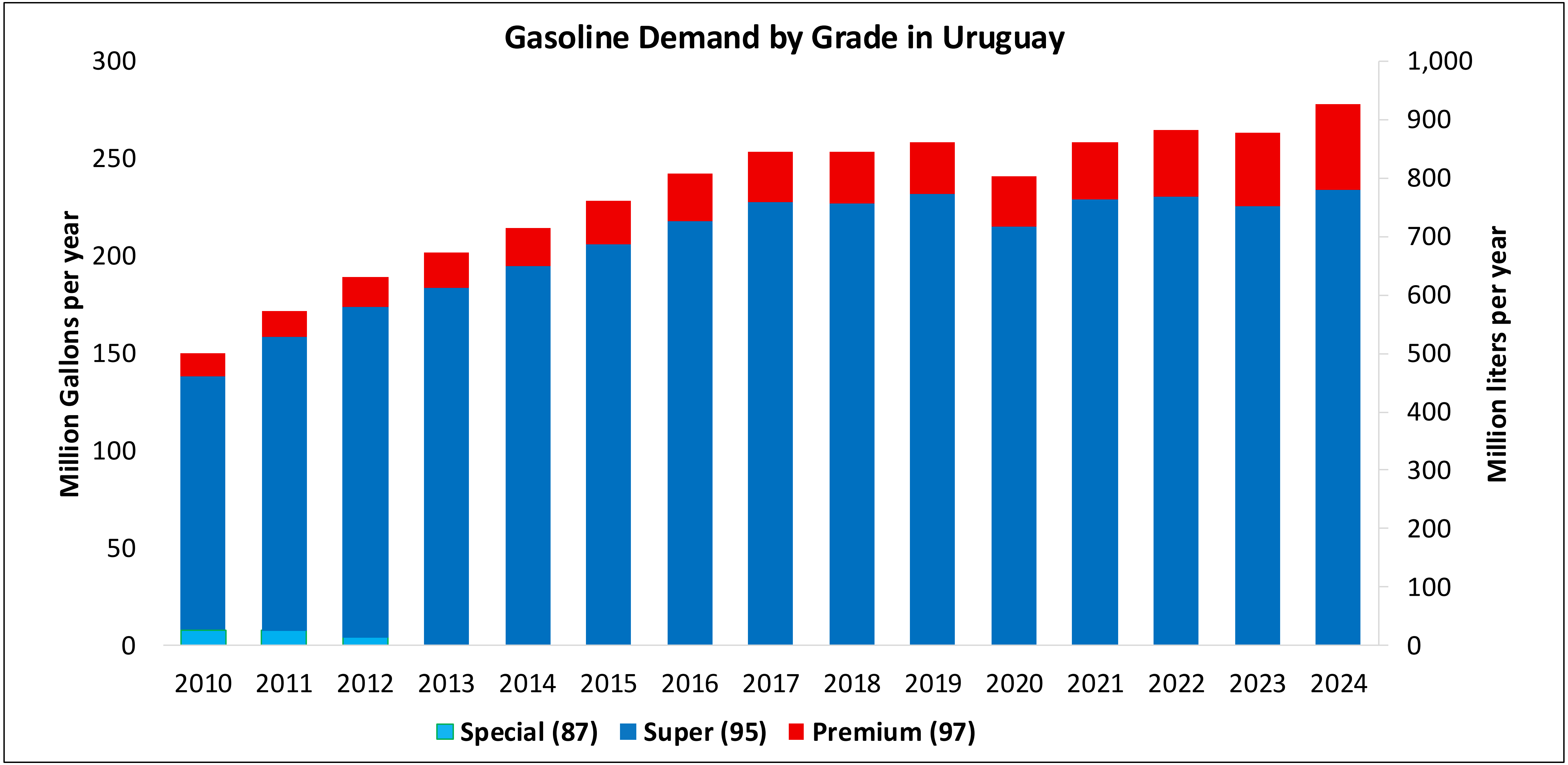
Ethanol Blending in Gasoline - Uruguay

In 2024, gasoline demand was 245 million gallons (926 million liters). Super Gasoline with RON 95 represented 84% of total demand and Premium Gasoline with RON 97 the rest. Gasoline production was 125 million gallons (471 million liters) and covers 64% of local demand.

Gasoline Blends with up to 10% v/v are allowed.. In Uruguay there are two ethanol plants with an overall capacity of 100 million liters per year, raw material commonly used are sugar cane, sorghum, and corn. In 2024, ethanol consumption was 24 million gallons (91 million liters).

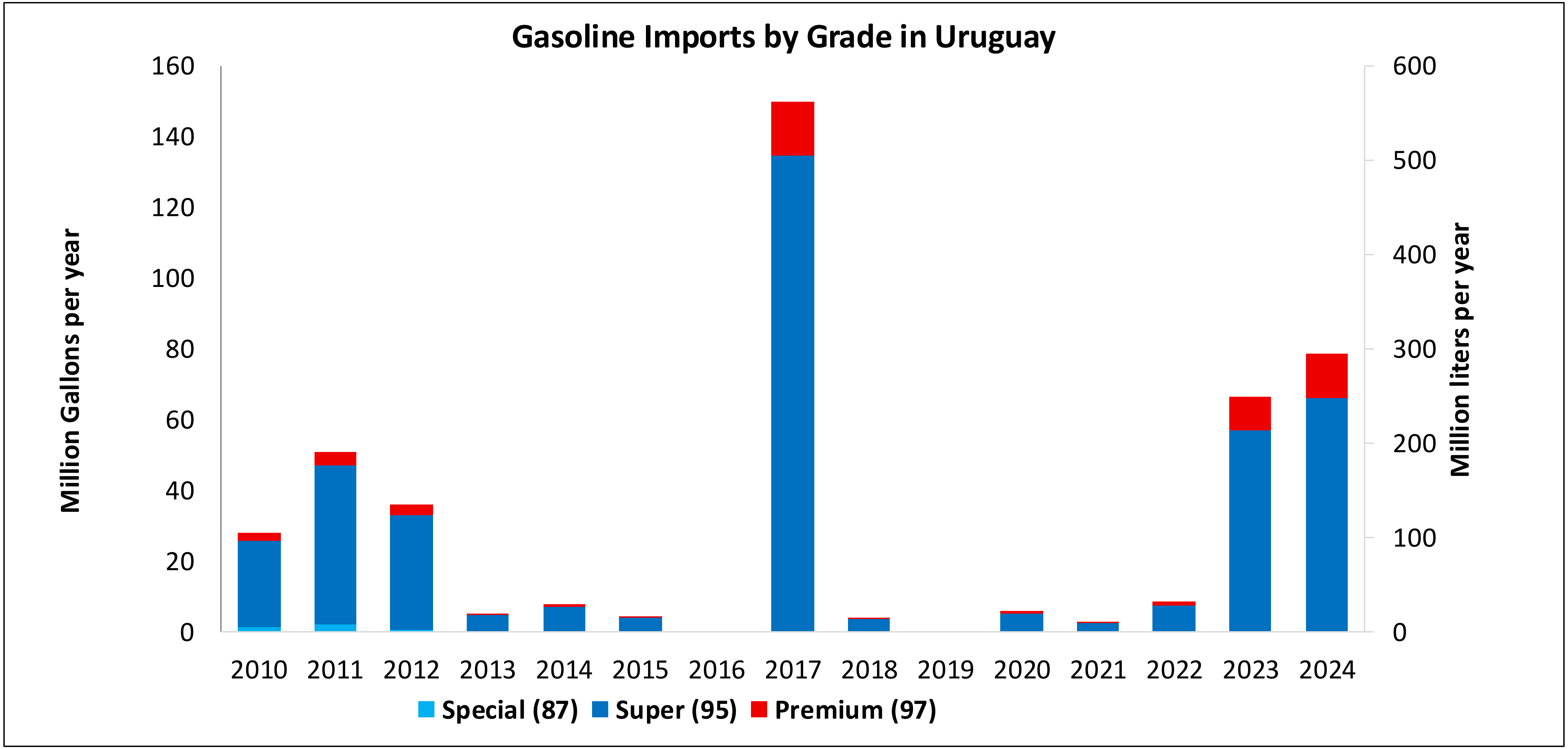
Source: ANCAP

Gasoline Demand in Uruguay



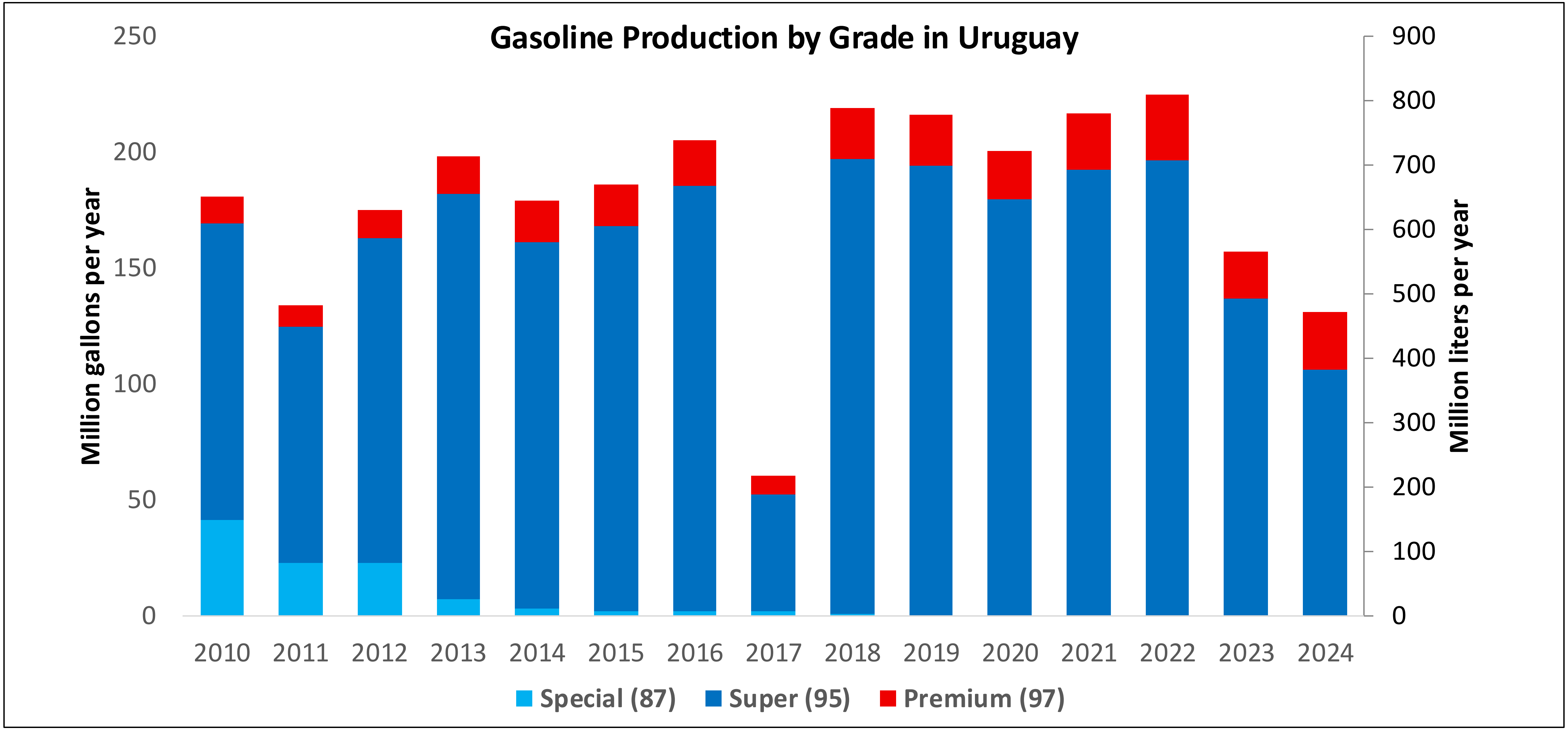
Source: ANCAP

Gasoline Imports in Uruguay



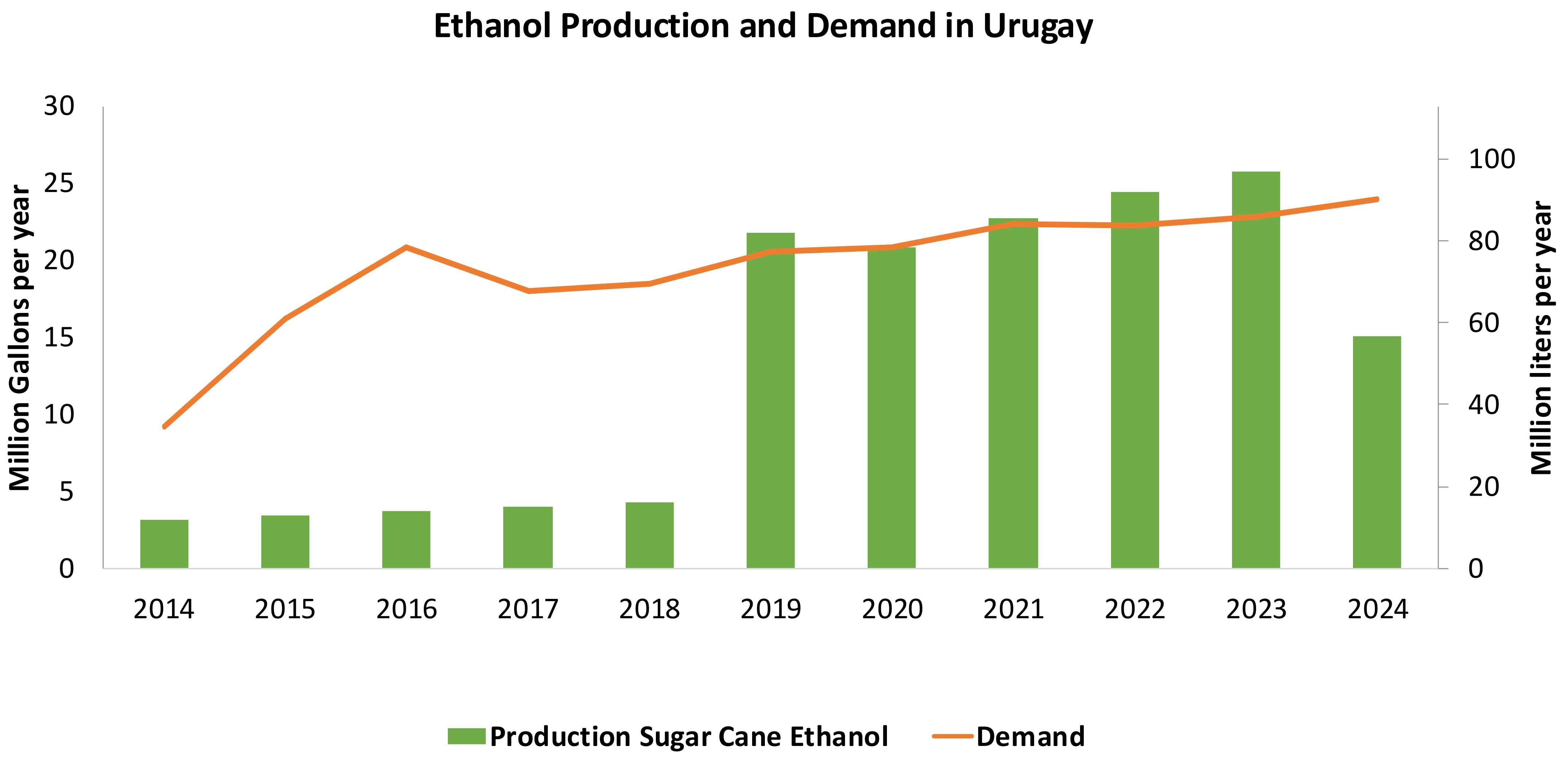
Source: ANCAP

Gasoline Production in Uruguay



Source: ANCAP

Ethanol Demand in Uruguay



Source: ANCAP

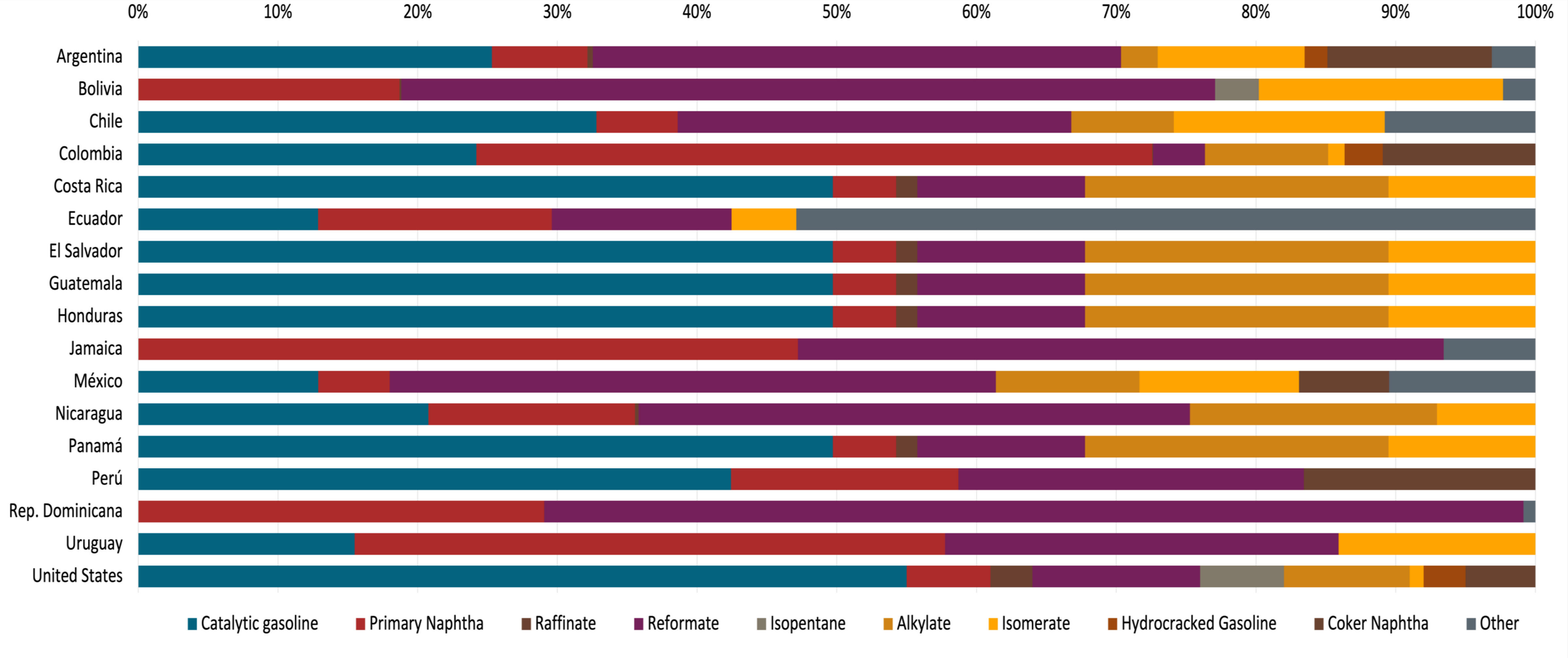
Gasoline Quality in Uruguay

Name	Resolution 110/2014		ANCAP		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2020		2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Super 95 30-S	Premium 97 30-S	Super 95 30-S	Premium 97 30-S	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 20 %v/v	< 20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 97	> 95	> 97	> 95	> 95	> 98	> 98
MON	> 82	> 84	> 85	> 85	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 30 mg/kg	< 30 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< > 10 %v/v	< > 10 %v/v			<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 72 kPa	<> 72 kPa	<> 50-80 kPa	<> 50-80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 83 kPa	<> 83 kPa	<> 45-67 kPa	<> 45-67 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: ANCAP

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



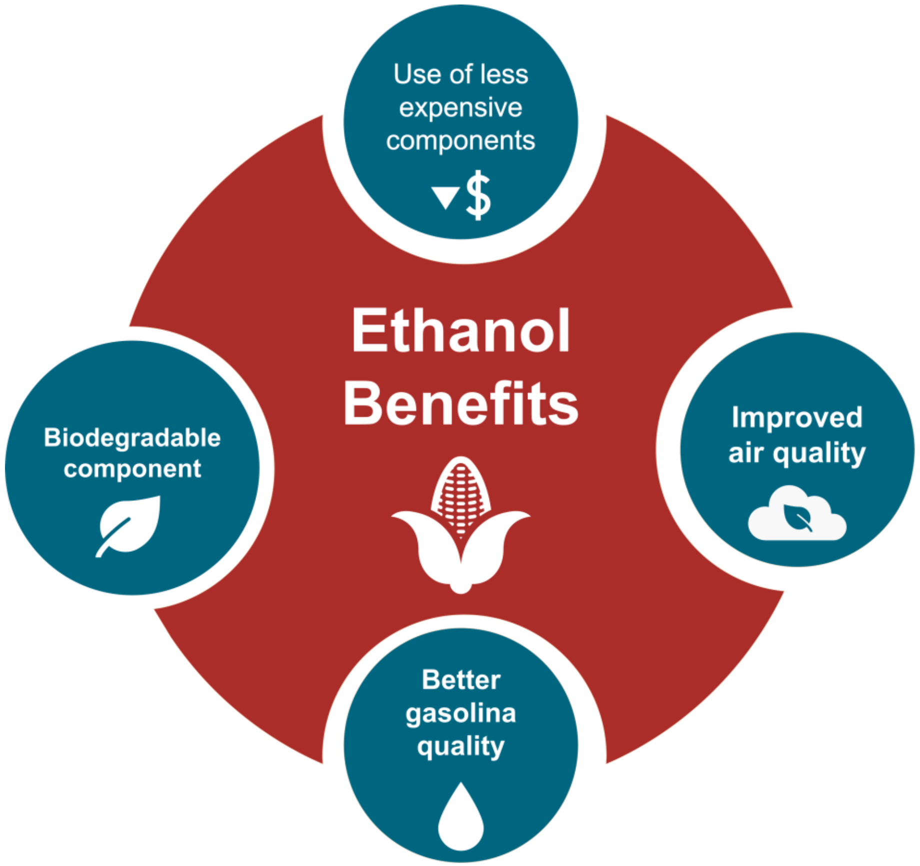
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

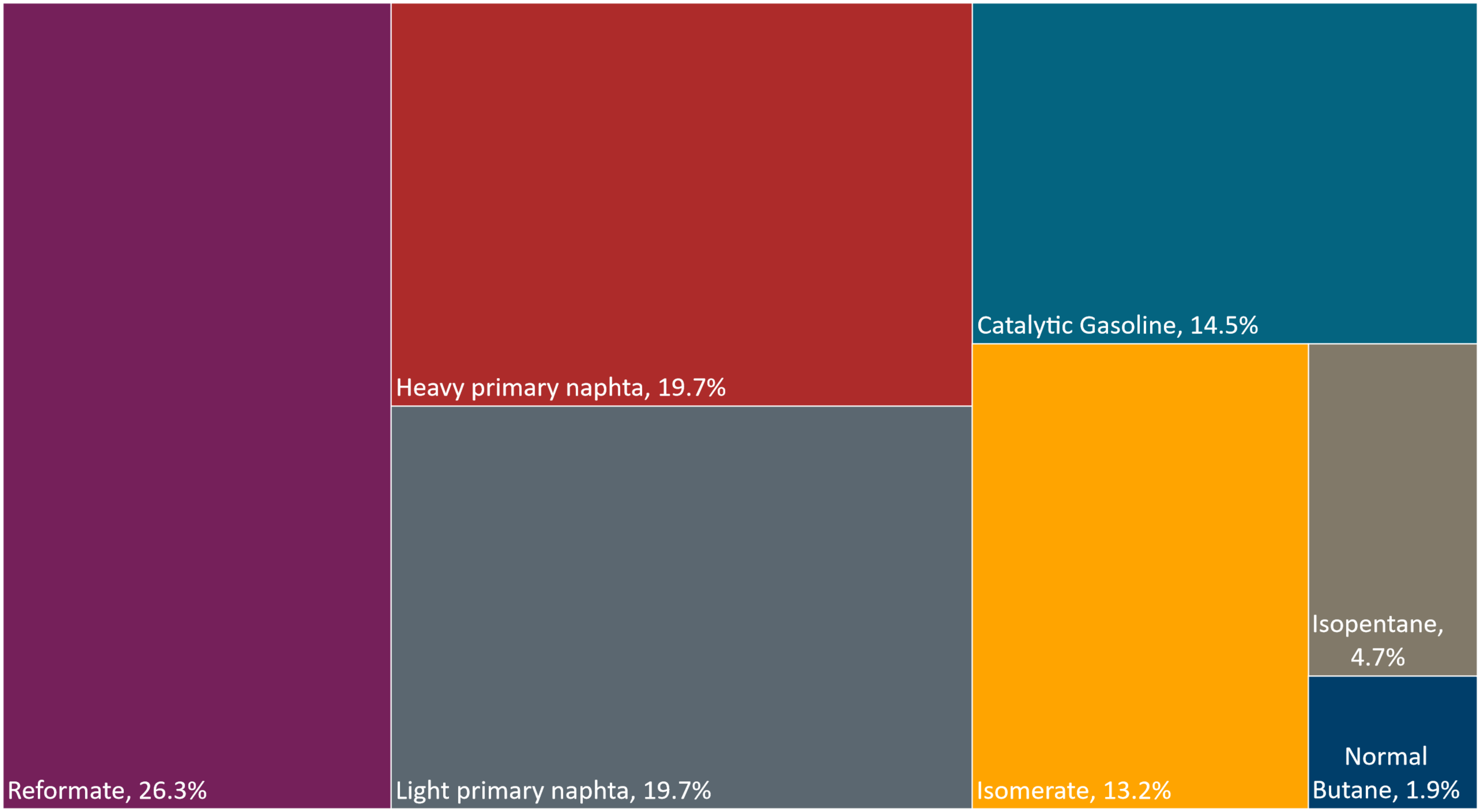
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average 2024 prices and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

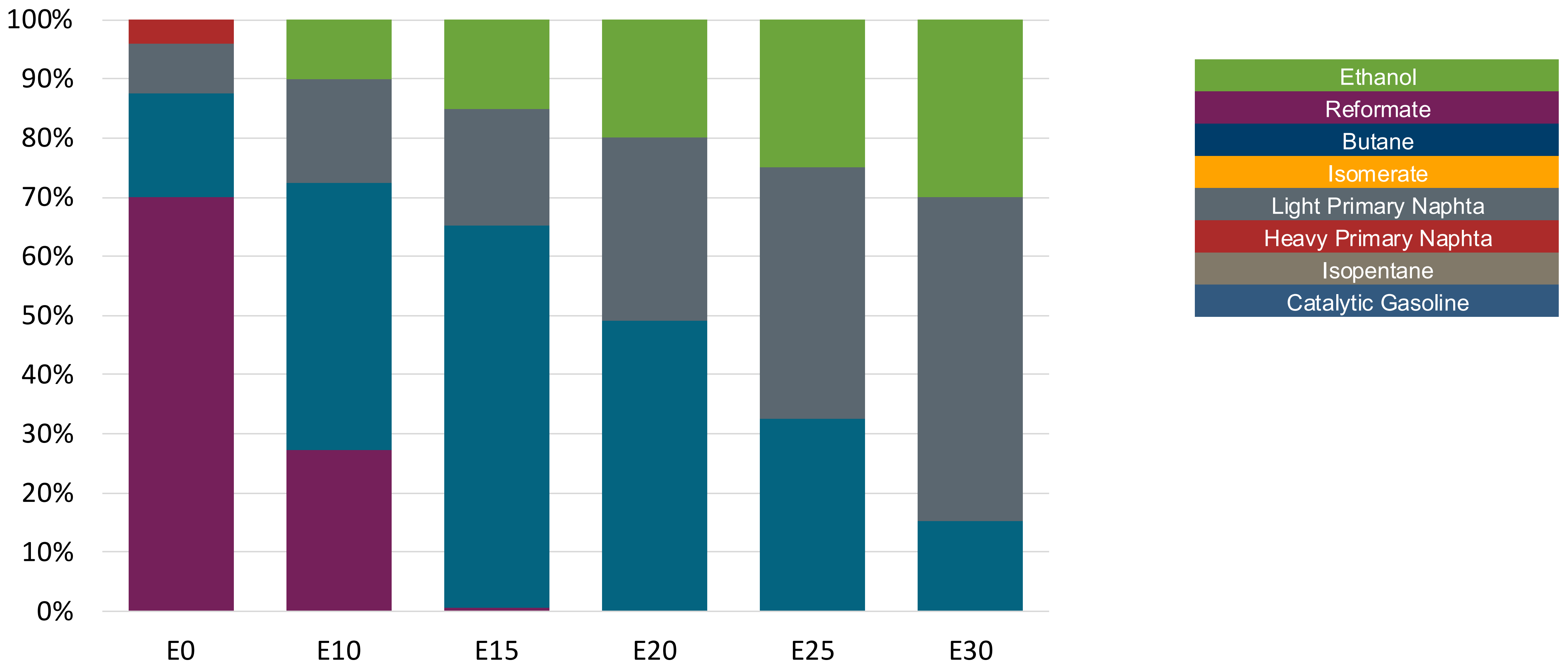


Available Blending Components



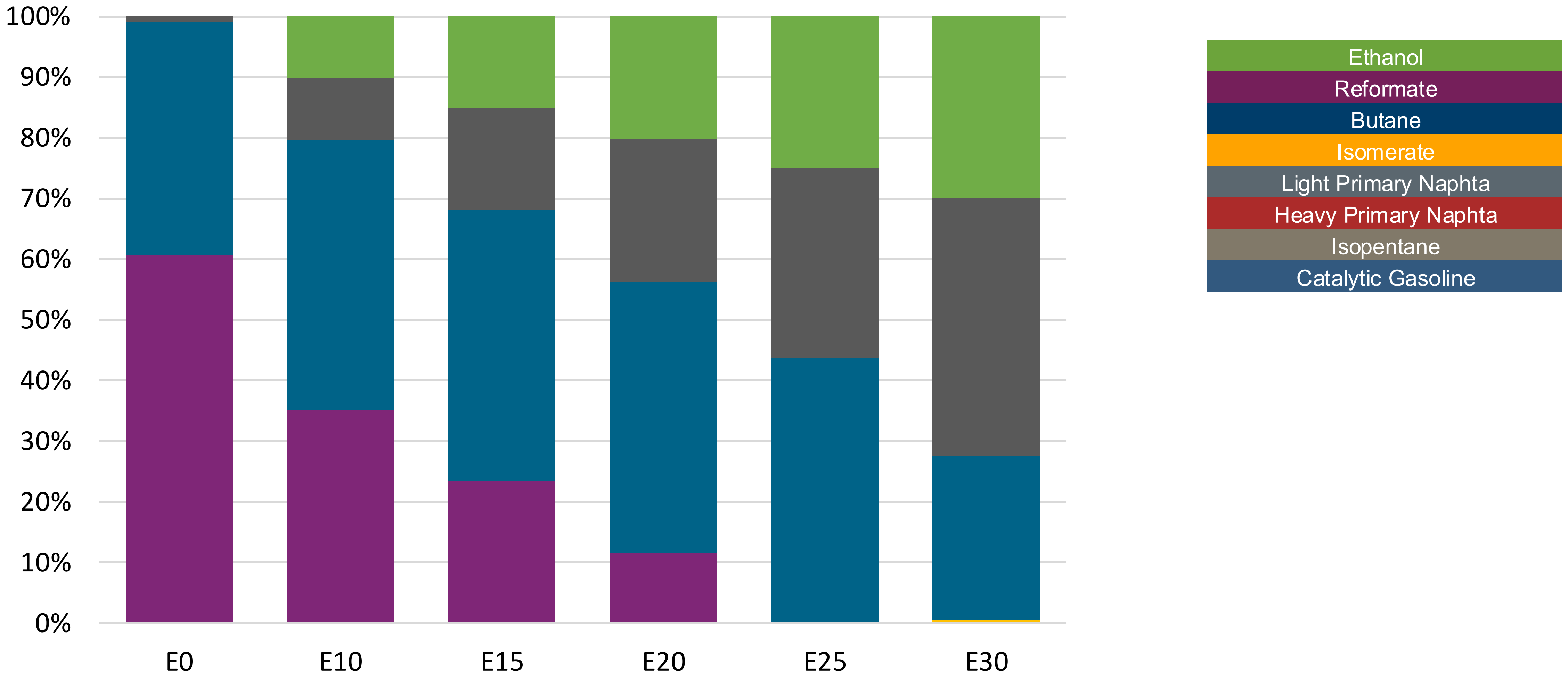
Source: Faro90

Super Gasoline – Constant Octane



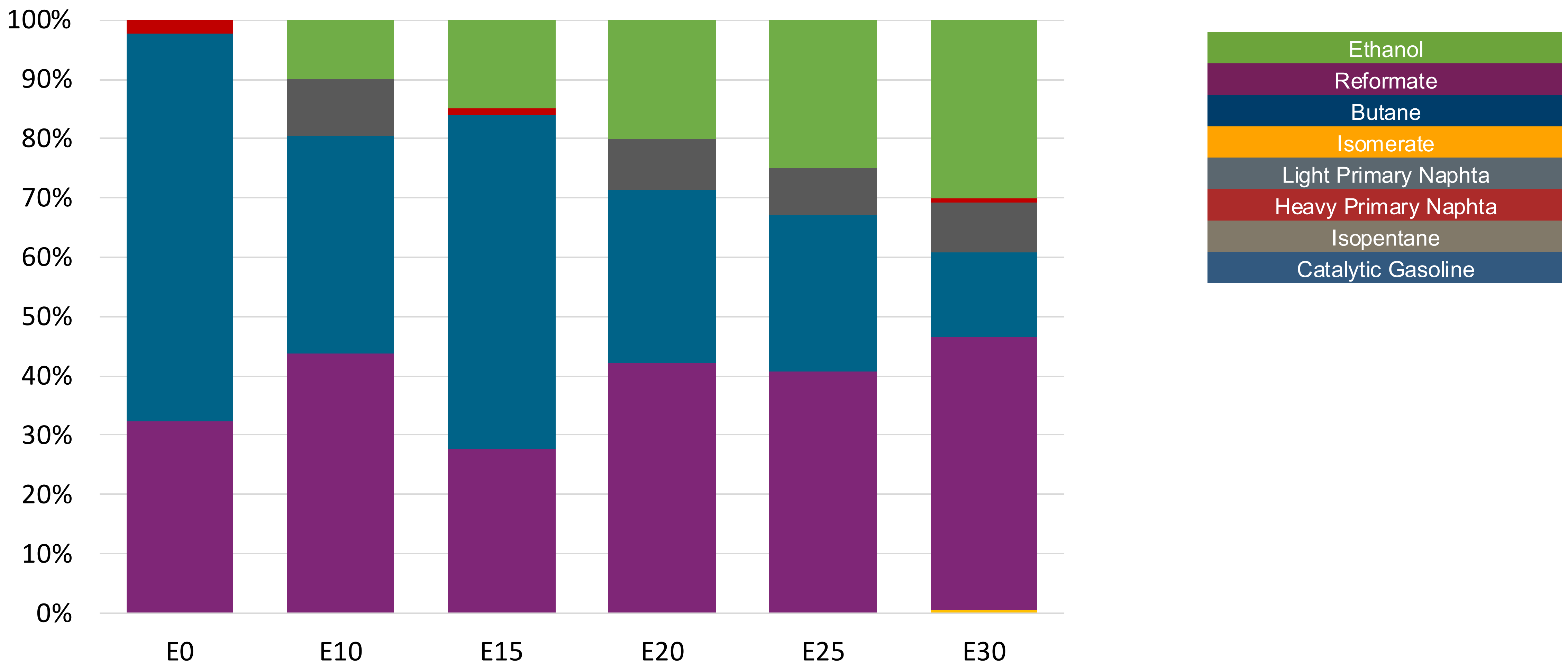
	E0	E10	E15	E20	E25	E30
Octane (RON)	94.7	95.5	95.5	95.5	95.5	95.5
Price (US$/gal)	$ 2.433	$ 2.239	$ 2.109	$ 2.012	$ 1.912	$ 1.807

Premium Gasoline – Constant Octane



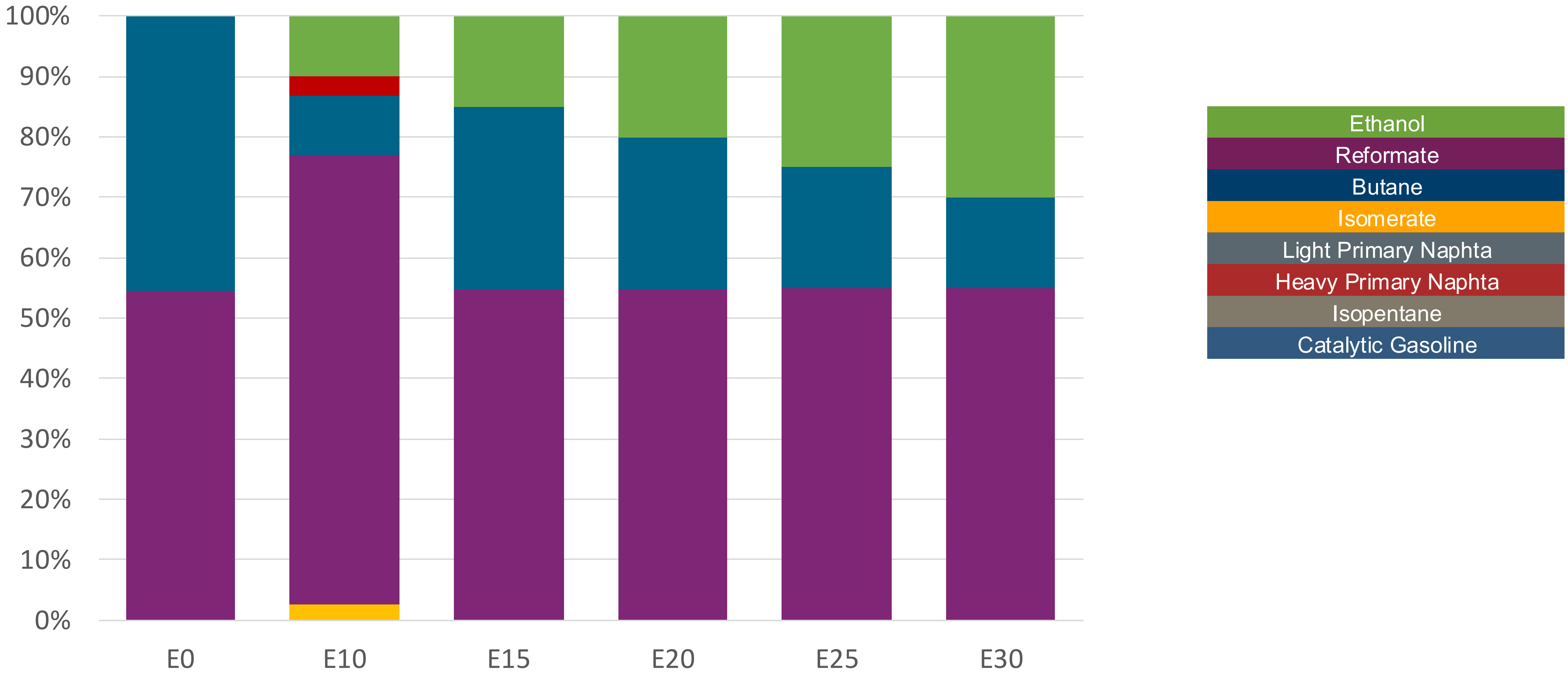
Octane (RON)	97.0	97.5	97.5	97.5	97.5	97.5
Price (US$/gal)	$ 2.519	$ 2.333	$ 2.229	$ 2.120	$ 2.008	$ 1.907

Super Gasoline – Increased Octane



Octane (RON)	94.5	98.2	100.8	102.1	103.9	105.5
Price (US$/gal)	$ 2.375	$ 2.375	$ 2.375	$ 2.375	$ 2.375	$ 2.375

Premium Gasoline – Increased Octane



Octane (RON)	96.8	100.5	102.7	104.5	106.2	107.7
Price (US$/gal)	$ 2.501	$ 2.501	$ 2.501	$ 2.501	$ 2.501	$ 2.501

Vehicle Emission Impact for Ethanol Gasoline Blending

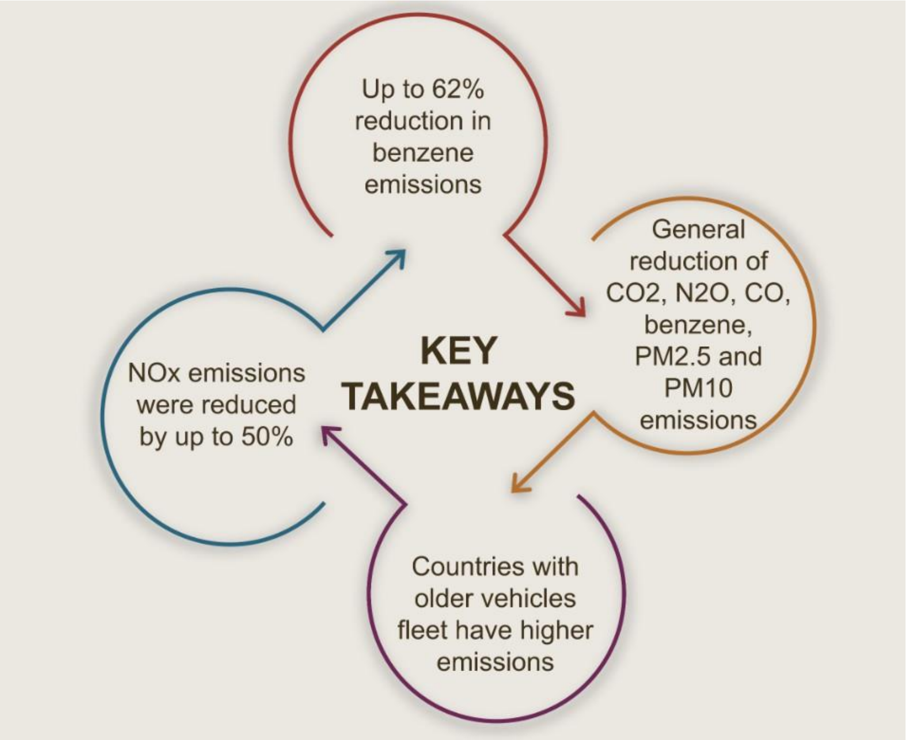
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

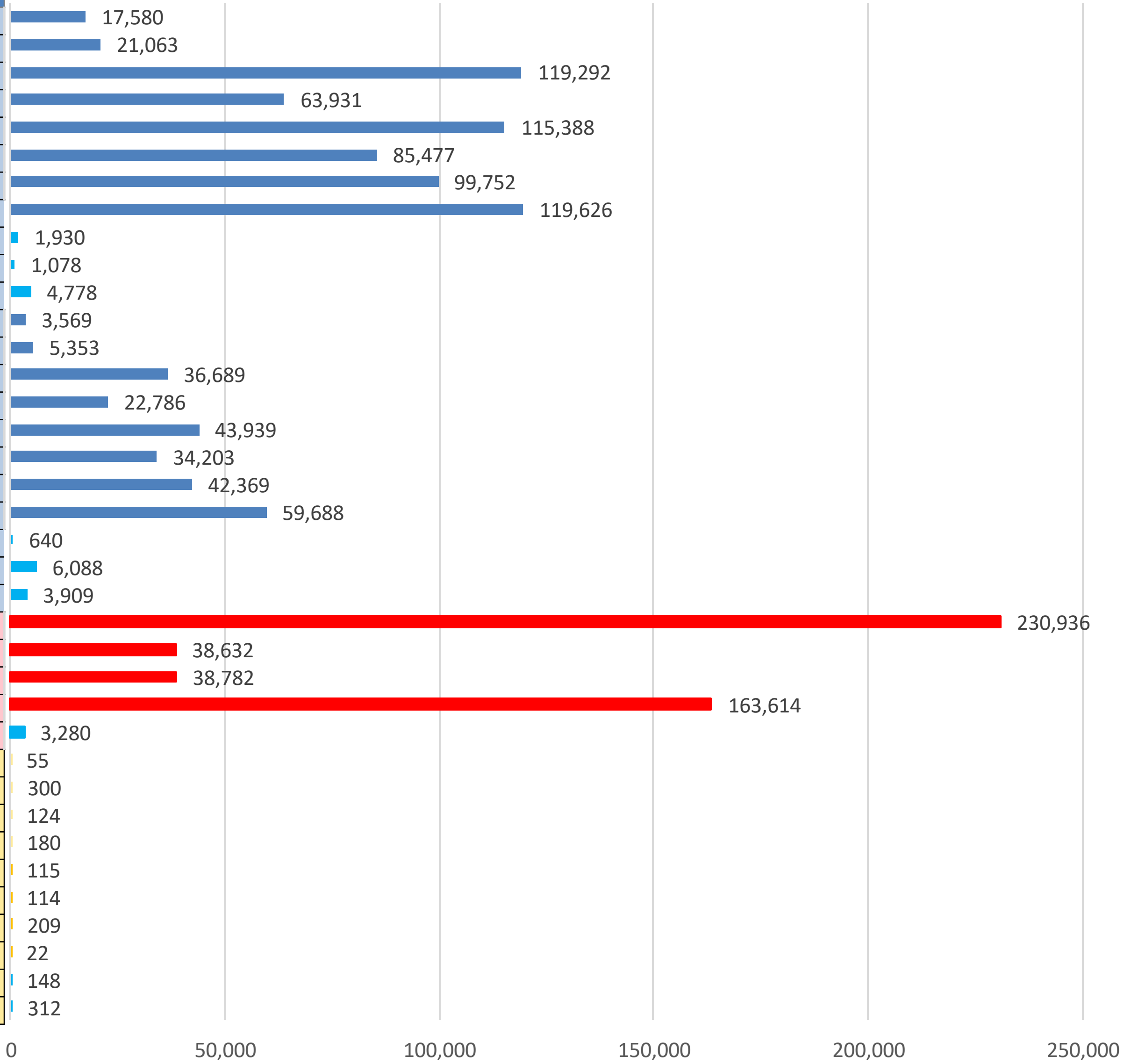
Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicular Fleet in Uruguay

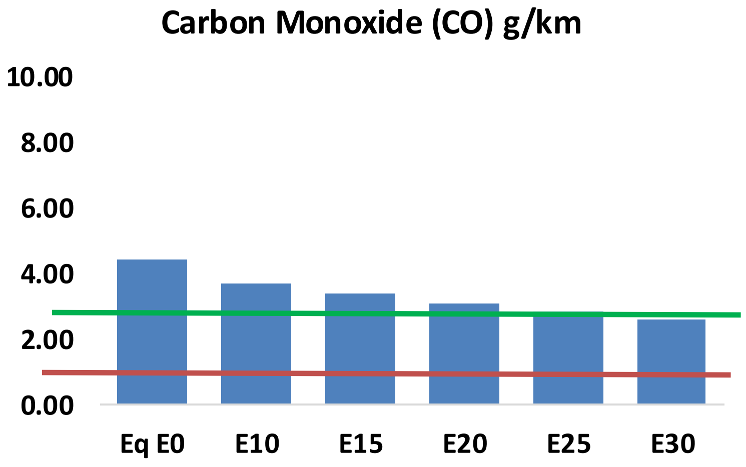
Type	Weight	Motor	Exhaust	Recuperator	Age
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
	Medium	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
Sml Engines	Light	4-Cycle	Euro II	PCV	+8years
					4-8years
			Euro III		4-8years
					-4years
		Eléctricas			-8years
Truck	Heavy	Carburator	3-Ways	PCV	+8years
		Fuel-Injection	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					4years
		Electric			

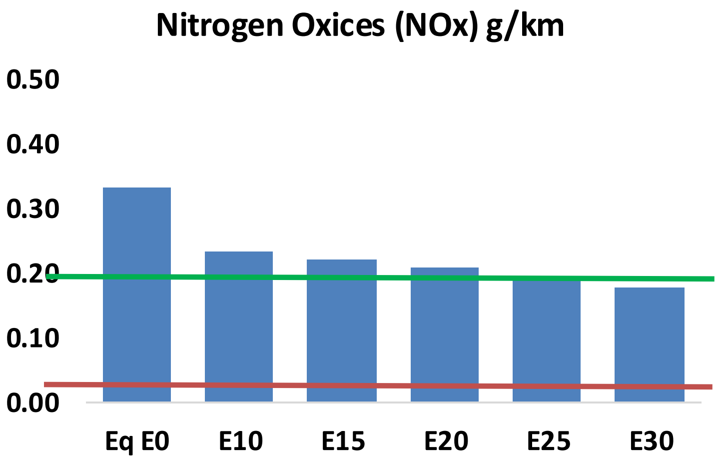


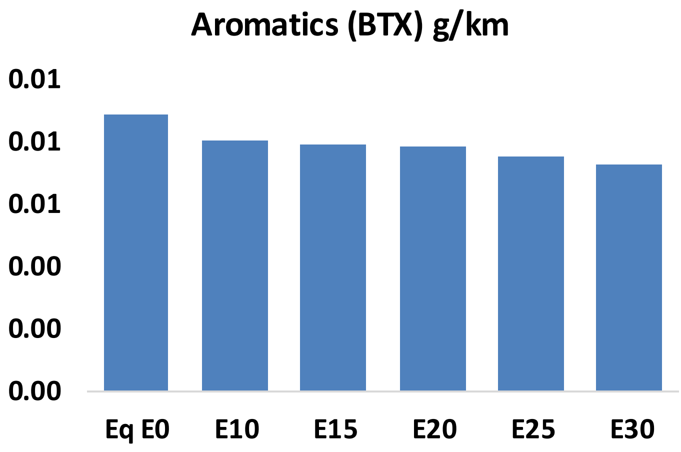
Gasoline Vehicular Fleet: **1,373,674**
Average Age: **8 years**
Source: MIEM, 2025.

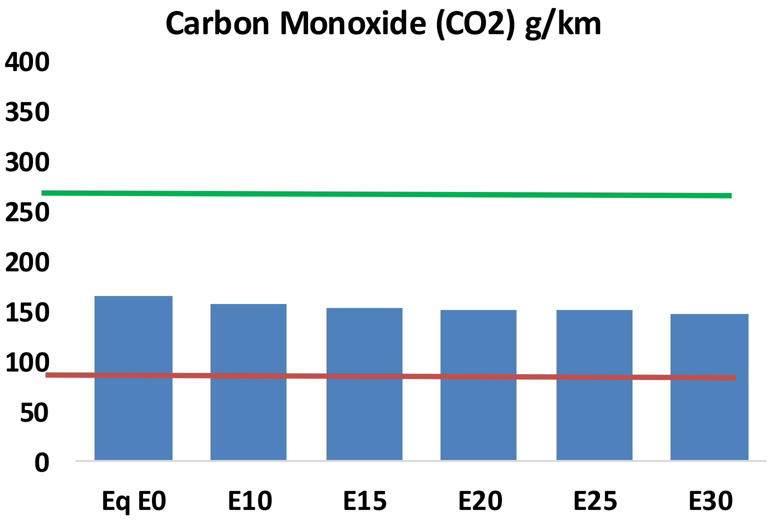
Vehicular Emissions in Uruguay

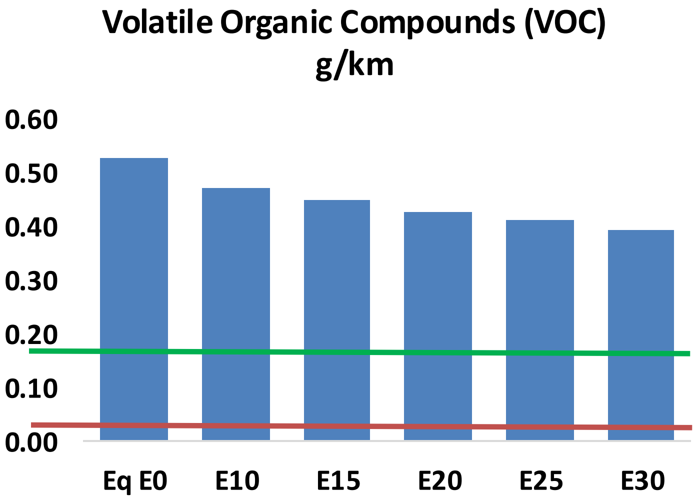
United States
Euro 6

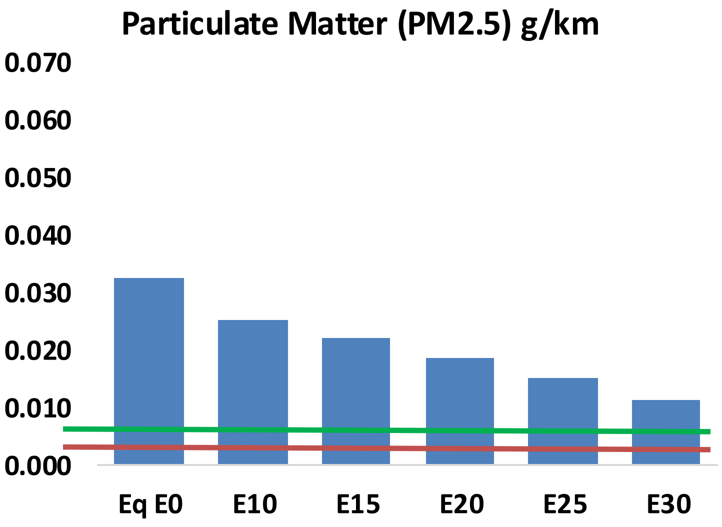












Source: Faro90, 2025.

Vehicular Emissions in Uruguay

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	151,843	139,219	126,595	116,064	105,382	97,142	88,035	-17%	-31%
CO2	5,613,763	5,469,409	5,333,075	5,225,852	5,172,794	5,145,685	5,050,835	-5%	-8%
VOC	17,961	16,985	16,008	15,286	14,596	14,076	13,344	-11%	-19%
NOx	11,383	8,537	7,968	7,510	7,083	6,611	6,113	-30%	-38%
SOx	66	61	56	52	48	44	39	-15%	-28%
NH3	1,925	1,906	1,887	1,896	1,917	1,932	1,945	-2%	0%
N2O	86	85	85	85	87	88	89	-1%	2%
CH4	4,530	4,530	4,530	4,597	4,697	4,783	4,865	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	138	132	126	122	118	115	111	-9%	-14%
Acetaldehyde	345	396	581	884	1,205	1,377	1,627	68%	249%
Formaldehyde	1,232	1,198	1,396	1,640	1,718	1,900	2,069	13%	39%
Benzene	301	290	274	270	268	256	248	-9%	-11%
PM2.5	373	331	289	251	213	173	131	-22%	-43%
PM10	733	651	568	494	419	340	258	-22%	-43%

Source: Faro 90, 2025.

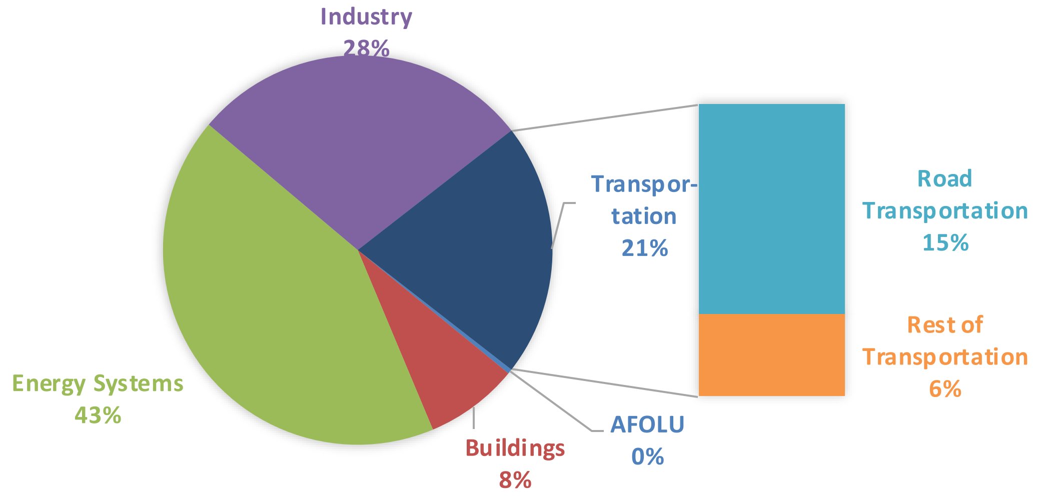
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

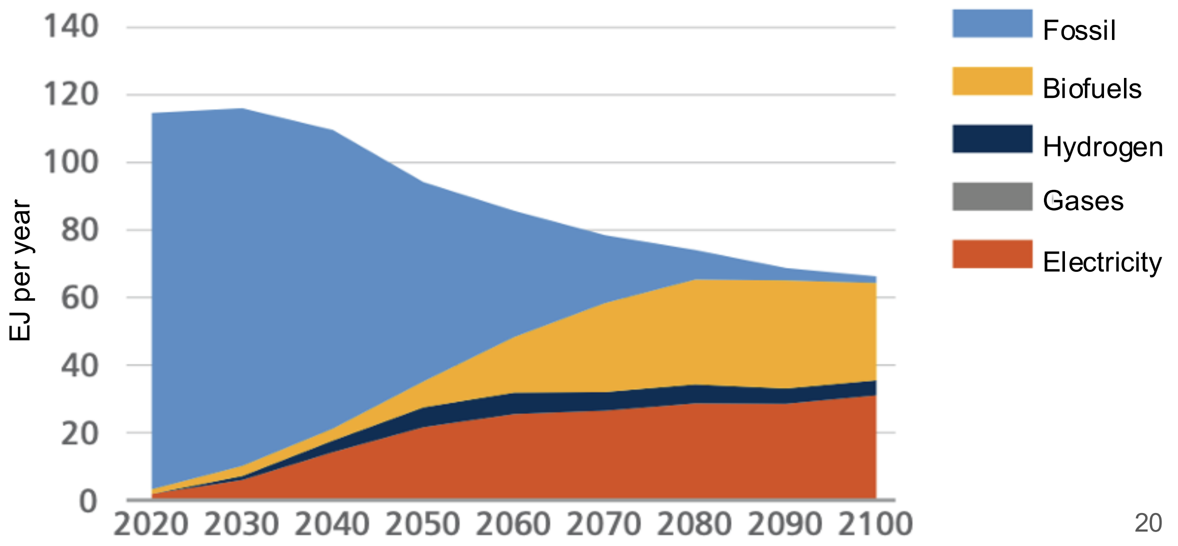
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

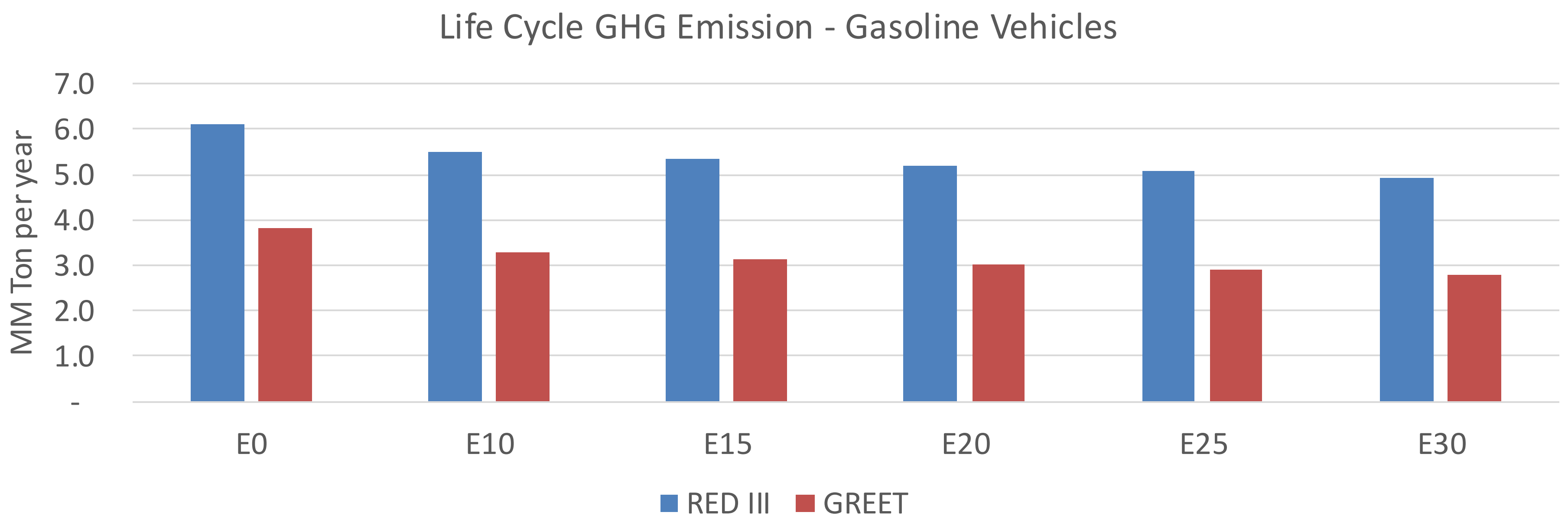
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Uruguay – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	44.6
2035 Target (MMT/year)	31.2
Delta	13.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.7%	18.2%	21.1%	24.0%	27.7%
RED II/III	0.0%	10.1%	12.7%	14.9%	17.1%	19.7%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.2%	5.2%	6.1%	6.9%	7.9%
RED II/III	0.0%	4.6%	5.8%	6.8%	7.8%	9.0%

Source: Greet, RED II/III, climateactiontracker.org, F90